

DOCUMENT: Research Guide — Topic 1

Antimicrobial Peptide (AMP) Activity & Toxicity Prediction with Protein Language Models

1. Motivation and Problem Statement

Antimicrobial resistance (AMR) has been declared a global health emergency by the WHO, and traditional wet-lab screening of antimicrobial peptides (AMPs) is slow, expensive, and low-throughput, creating strong demand for reliable computational pre-screening tools. This project will build a machine learning pipeline that predicts both antimicrobial **activity** and **toxicity/hemolysis** of peptide sequences, closing a gap that most existing single-task models leave open. <https://pubs.acs.org/doi/10.1021/acs.jcim.5c00006>, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11756916/>, <https://pubmed.ncbi.nlm.nih.gov/40626654/> (Deep Learning in Antimicrobial Peptide Prediction)

2. Biological Background Required

- Amino acid properties: charge, hydrophobicity, polarity
- Peptide bonds and primary sequence structure
- What makes a peptide "antimicrobial" (amphipathicity, membrane disruption mechanism)
- Basic concept of hemolysis and cytotoxicity as a safety constraint
- Overview of antimicrobial resistance and why AMPs are a proposed alternative to antibiotics. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11756916/>

3. ML/DL Background Required

- Sequence embeddings and tokenization
- Transformer architecture fundamentals
- Protein language models (ESM-2) and transfer learning
- Binary and multi-label/multi-task classification
- Class imbalance handling (AMP datasets are typically imbalanced)
- Basics of interpretability (SHAP, attention visualization)

4. Datasets

Dataset	Description	Link
DBAASP	Curated AMP database with activity/MIC values against specific pathogens	dbaasp.org
APD3 (Antimicrobial Peptide Database)	Classic curated AMP repository with sequence + activity annotations	aps.unmc.edu/AP
DRAMP	Large repository including natural, synthetic, and patent AMPs with structured labels	dramp.cpu-bioinform.org

All three are freely downloadable in FASTA/CSV format and require no credentialing.pubmed.ncbi.nlm.nih/1

5. Recent SOTA Methods

- **deep-AMPPred**: ESM-2 encoding combined with CNN, BiLSTM, and CBAM attention for AMP identification and multi-label functional activity prediction (JCIM, 2025). [link](#)
- **BERT-AmPEP60**: Fine-tuned BERT-based transfer learning model for MIC regression against *E. coli* and *S. aureus*, outperforming classical ML/DL baselines (JCIM, 2025) [Article](#) [CodeSource](#), [Article](#)
- Generative AI pipelines for de novo AMP discovery against multidrug-resistant bacteria (Nature Microbiology, 2025) [nature](#)

6. Classic Baseline Methods

- SVM / Random Forest on handcrafted descriptors: Amino Acid Composition (AAC), Composition-Transition-Distribution (CTD), physicochemical descriptors
- CNN / LSTM on one-hot or BLOSUM-encoded sequences
- AMP-EBiLSTM (BiLSTM-based baseline) [frontiersin](#)

7. Foundation Models

- **ESM-2** (Meta AI) — primary embedding source, use the small variant (35M or 150M parameters) to fit Colab GPU limits
- **ProtBERT** — alternative protein language model for comparison
- **ESMFold** — optional, for structure-aware feature extraction if time permits

8. Key Papers to Read (minimum coverage)

5 key papers:

1. deep-AMPPred — ESM-2 + CNN/BiLSTM/CBAM for AMP identification and functional activity. [pubs.acs](#)
2. BERT-AmPEP60 — BERT-based transfer learning for MIC regression [pubs.acs+1](#)
3. AMP-EBiLSTM — BiLSTM baseline model. [frontiersin](#)
4. Classic AMP review (Frontiers in Genetics, 2023). [frontiersin](#)
5. AMP database benchmarking review (Briefings in Bioinformatics, 2023). [academic.oup](#)

3 recent SOTA papers (2023–2026):

1. Deep Learning for AMPs: Computational Models and Databases (JCIM, 2025) [pubs.acs](#)
2. Generative AI approach for AMP discovery against multidrug-resistant bacteria (Nature Microbiology, 2025) [nature](#)
3. Recent advances in computational AMP discovery (Frontiers in Bioinformatics, 2026) [frontiersin](#)

2 survey/review papers:

1. Deep Learning for Antimicrobial Peptides: Computational Models and Databases (JCIM, 2025) [pubs.acs](#)
2. Machine Learning for AMP identification and design (Nature Reviews Bioengineering, 2024) [nature](#)

9. What Is Missing in the Literature

Most existing models treat activity, toxicity, and hemolysis as separate single-task problems rather than jointly modeling them, and few compare ESM-2 embeddings fairly against classical descriptors under identical experimental conditions. Interpretability remains weak — very few papers link learned attention patterns back to known biological motifs (e.g., amphipathic helix regions). [pubmed.ncbi.nlm.nih](#)

Etc..

10. Possible Research Gaps

- Lack of unified multi-task benchmarks combining activity + toxicity + hemolysis prediction
- No standardized fair comparison protocol between embedding-based and feature-based approaches
- Weak cross-database generalization testing (models trained on DBAASP rarely tested on DRAMP or APD3)
- Limited interpretability studies connecting attention weights to biologically known motifs
- ETC ...

11. Suggested Original Contributions

1. A reproducible benchmark comparing ESM-2 embeddings vs. classical AAC/CTD descriptors on the same splits
2. A multi-task classifier jointly predicting activity and toxicity/hemolysis
3. A SHAP/attention-based interpretability study linking model decisions to known biological motifs
4. A data augmentation strategy via point mutations to address class imbalance
5. A small, novel cross-database benchmark (train on DBAASP, test on DRAMP/APD3) to assess generalization
6. Etc...

12. Evaluation Metrics

Accuracy, AUC-ROC, Matthews Correlation Coefficient (MCC), F1-score, and Precision-Recall AUC (critical given severe class imbalance in AMP datasets). etc

13. Experimental Protocol

Data cleaning and deduplication → embedding extraction (ESM-2) → classical baseline training (SVM/RF) → deep learning classifier training → ablation study (handcrafted features vs. embeddings) → multi-task extension → interpretability analysis (SHAP/attention) → cross-database generalization test.

14. Computational Resources

A single GPU is sufficient (Google Colab T4 or better); use the ESM-2 small variant (35M or 150M parameters) to keep training times manageable within the 6-week window.

15. Target Venues

- **Journal of Chemical Information and Modeling (JCIM)** — accepts ML models on chemical/biological data; requires strong experimental validation, not purely computational docking [pubs.acs](https://pubs.acs.org/journal/jcim) (recommended: you can read papers from this journal)
- **Frontiers in Bioinformatics**
- **IEEE/ACM BIBM workshops** (faster turnaround, good first venue for undergraduate work)

What each target journal expects (to plan for from Week 1):

- **JCIM:** strong methodological novelty, comparison against SOTA, and validation beyond simple accuracy (statistical significance testing is expected)
- **BMC Bioinformatics:** full reproducibility — public dataset links, open-source code repository, clear "Availability of data and materials" section, and a cover letter explaining the manuscript's specific contribution.
- ...

Advice

Do the literature review seriously in Week 1 — a shallow state-of-the-art review is the single biggest reason projects fail to find a genuine contribution. Read not just abstracts but methods and limitations sections; every "Limitations" or "Future Work" paragraph in a SOTA paper is a potential contribution for you. Keep a shared table of "paper → method → dataset → reported metric → limitation" from day one; this becomes your related-work section and your contribution justification simultaneously. Do not skip the baseline, reviewers judge novelty relative to baselines, not just SOTA. Push for reproducibility discipline (fixed seeds, documented splits) from Week 2 onward, since irreproducible results are the top rejection reason at Q2/Q3 venues.

Six-Week Detailed Roadmap

Week 1 — Foundations & Literature Review

- **Learning objectives:** Understand AMP biology, the AMR problem, and how protein language models represent sequences
- **Biology to study:** Amino acid properties, peptide bonds, amphipathicity, mechanism of AMP action, AMR background. [Link](#)

- **AI/ML to study:** Transformer basics, ESM-2 architecture, transfer learning concepts
- **Papers to read:** Survey, Survey, classic reviewnature+2
- **Coding tasks:** Install Biopython, fair-esm/HuggingFace ESM-2, set up Python environment
- **Dataset preparation:** Download DBAASP, DRAMP, APD3; inspect label formats and overlaps
- **Experiments:** None (exploration phase)
- **Deliverables:** Literature review table (paper/method/dataset/metric/limitation) uploaded to Drive; environment setup confirmed
- **Milestone:** Working environment + dataset understanding validated in supervision meeting

Week 2 — Preprocessing & Baseline

- **Learning objectives:** Build a clean, reproducible data pipeline and a working classical baseline
- **Biology to study:** AAC/CTD descriptor meaning, physicochemical descriptor interpretation
- **AI/ML to study:** SVM/Random Forest fundamentals, stratified train/val/test splitting for biological sequences
- **Papers to read:** AMP-EBiLSTM baseline paper. [frontiersin](#)
- **Coding tasks:** Preprocessing pipeline, AAC/CTD feature extraction, SVM/RF baseline implementation
- **Dataset preparation:** Deduplicate, clean, stratified splits (train/val/test)
- **Experiments:** Baseline SVM/RF classification results
- **Deliverables:** Cleaned dataset (CSV) + preprocessing notebook on Drive; data/ and notebooks/01_preprocessing.ipynb on GitHub
- **Milestone:** Reproducible baseline pipeline running end-to-end

Week 3 — SOTA Reproduction

- **Learning objectives:** Reproduce a recent SOTA model and critically evaluate its limitations
- **Biology to study:** Functional AMP activity classes, MIC concept. [research.mpu](#)
- **AI/ML to study:** ESM-2 embedding extraction, fine-tuning strategy, MCC/AUC/F1 evaluation
- **Papers to read:** deep-AMPpred, BERT-AmPEP60
- **Coding tasks:** ESM-2 embedding extraction + classifier head; reproduce reported SOTA numbers
- **Dataset preparation:** Verify embedding quality via UMAP/clustering visualization
- **Experiments:** SOTA reproduction vs. baseline comparison
- **Deliverables:** SOTA reproduction report on Drive; notebooks/02_sota_reproduction.ipynb + results CSV on GitHub
- **Milestone:** SOTA performance reproduced within acceptable margin; limitations identified

Week 4 — Designing the Original Contribution

- **Learning objectives:** Design and implement a genuine improvement over reproduced SOTA
- **Biology to study:** Hemolysis/toxicity biology, cross-species AMP activity variability
- **AI/ML to study:** Multi-task learning architectures, attention-based interpretability
- **Papers to read:** Generative AI AMP discovery paper; one student-sourced paper on multi-task peptide models. [nature](#)
- **Coding tasks:** Implement multi-task head (activity + toxicity); implement embedding-vs-feature ablation script etc
- **Dataset preparation:** Prepare toxicity/hemolysis-labeled subset
- **Experiments:** Ablation (embeddings vs. handcrafted features); multi-task vs. single-task comparison etc
- **Deliverables:** Ablation results + interpretability plots on Drive; experiments/ablation/ on GitHub
- **Milestone:** Original model shows measurable improvement or new insight vs. baseline/SOTA

Week 5 — Large-Scale Evaluation & Explainability

- **Learning objectives:** Validate results rigorously and produce publication-grade analysis
- **Biology to study:** Motif-level biological validation of predicted important regions
- **AI/ML to study:** SHAP/attention visualization, bootstrap/statistical significance testing
- **Papers to read:** Recent advances review (skim for context on remaining SOTA comparisons) [frontiersin](#)
- **Coding tasks:** Large-scale experiments, statistical significance testing, cross-database generalization test, error analysis by peptide family
- **Dataset preparation:** Cross-database test sets (train DBAASP → test DRAMP/APD3)
- **Experiments:** Full SOTA comparison table, generalization test, error analysis
- **Deliverables:** Final results tables/figures (high-res) on Drive; results/final/ folder on GitHub
- **Milestone:** All experiments finalized, validated, and statistically supported

Week 6 — Writing & Publication Preparation

- **Learning objectives:** Produce a submission-ready manuscript and reproducible repository
- **AI/ML to study:** Scientific writing structure (IEEE/JCIM/BMC formats), reproducibility standards
- **Papers to read:** None new review target journal author guidelines
- **Coding tasks:** Generate publication-quality figures; clean GitHub repository (README, requirements.txt, LICENSE)
- **Dataset preparation:** Package supplementary data/code for release
- **Experiments:** None (finalize existing results only)
- **Deliverables:** Full paper draft (Overleaf link) covering Abstract, Introduction, Related Work, Methodology, Experiments, Discussion, Conclusion; supplementary material folder on Drive; tagged GitHub release v1.0

- **Milestone:** Submission-ready manuscript with identified target venues (JCIM, BMC Bioinformatics, Frontiers in Bioinformatics, or IEEE BIBM)